American Economic Journal: Applied Economics 2025, 17(1): 271–296
https://doi.org/10.1257/app.20230089

Staggered Difference-in-Differences in Gravity Settings: Revisiting the Effects of Trade Agreements[†]

By Arne J. Nagengast and Yoto V. Yotov*

We nest an extended two-way fixed effect (ETWFE) estimator for staggered difference-in-differences within the structural gravity model. To test the ETWFE, we estimate the effects of regional trade agreements (RTAs). The results suggest that RTA estimates in the current gravity literature may be biased downward (by more than 50 percent in our sample). Sensitivity analyses confirm the robustness of our main findings and demonstrate the applicability of our methods across settings. We expect the ETWFE methods to have significant implications for estimates of other policy variables in the trade literature and for gravity regressions on migration and foreign direct investment. (*JEL* C23, F14, F15, F16, J82)

We show that accounting for heterogeneous effects of regional trade agreements (RTAs) across space and time leads to estimates of RTA effects on trade that are twice as large as previously thought. To highlight our contribution, Figure 1 capitalizes on the analysis from Larch and Yotov (2023), who offer a chronological review of the developments in the RTA literature over the past 60 years; from the first atheoretical gravity specification of Tinbergen (1962), which is labeled "Tinbergen: 1962," until the most recent developments in the empirical trade literature, as captured by specification "Consecutive years: 2022," which comes from Egger, Larch, and Yotov (2022).[1]

* Nagengast: Deutsche Bundesbank (email: arne.nagengast@bundesbank.de). Yotov: Drexel University and ifo, CESifo (email: yotov@drexel.edu). Xavier Jaravel was coeditor for this article. We would like to thank Jeffrey Wooldridge and two anonymous referees for insightful comments that greatly improved the paper. We are also grateful to Sebastien Bradley, Peter Egger, Teresa Harrison, Ohyun Kwon, Christopher Meissner, Ninon Moreau-Kastler, Fernando Rios-Avila, Farid Toubal, Thomas Zylkin, and seminar participants at the Deutsche Bundesbank and the Economics of Free Trade Agreements conference at the University of Surrey for their helpful comments and suggestions. All remaining errors are our own. The views expressed in this paper are those of the author(s) and do not necessarily coincide with the views of the Deutsche Bundesbank or the Eurosystem.

† Go to https://doi.org/10.1257/app.20230089 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

[1] To construct Figure 1, we use the bilateral trade dataset of the World Trade Organization (Larch et al. 2019b), downloadable at https://www.wto.org/english/res_e/reser_e/structural_gravity_e.htm, and we implement some of the specifications from Larch and Yotov (2023). We offer further details on the sources and construction of our data in Section II. Specification "Consecutive years: 2022" incorporates the current best practice recommendations from the empirical trade literature, including the Poisson pseudo-maximum likelihood (PPML) estimator, exporter-time and importer-time fixed effects, pair fixed effects, domestic trade flows, globalization effects, consecutive-year data, and pair-clustered standard errors. We briefly review the current methods of estimating the gravity model in Section IA.

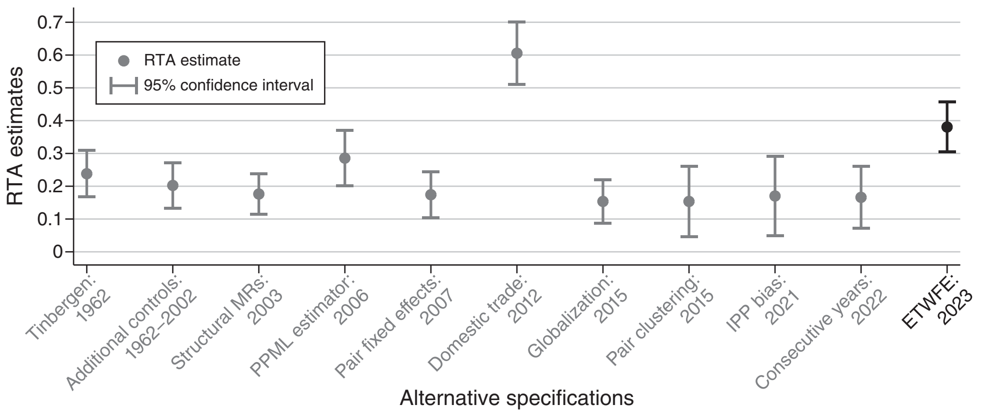

Figure 1. 61 Years of RTA Estimates with the Gravity Model of Trade

Notes: This figure replicates some of the specifications from figure 1 of Larch and Yotov (2023) with our data. The figure plots estimates of the effects of RTAs along with the corresponding 95 percent confidence intervals. The *x*-axis of the figure lists the alternative specifications from the existing literature, which follows the evolution of the estimation methods from the 1962 specification of "Tinbergen" to the most recent developments that are taken into account in specification "Consecutive years." The last estimate in the figure is obtained with the extended two-way fixed effect (ETWFE) estimator that we implement in this paper. See Section III for further details.

We draw three conclusions from Figure 1. First, from a methodological perspective, the main message is that there have been many and impactful developments in the related literature. Second, from a policy perspective, our estimates suggest that the effects of RTAs on trade have been positive and statistically significant but relatively small. The most current estimate (Consecutive years: 2022) implies that, all else equal, the RTAs in our sample have led to an increase in trade among the RTA members of about 18.1 percent.[2] Lastly, based on the evolution of the estimates in Figure 1, it seems that the existing literature has converged toward a stable RTA estimate of about 0.17–0.18.

Our main result appears last in Figure 1. The specification that we use to obtain it is identical to the preferred trade model (Consecutive years: 2022), but, in addition, it implements the latest heterogeneity-robust innovations in the difference-in-differences (DiD) literature on staggered treatment adoption, that is, when the treatment or intervention occurs in multiple units in different time periods (e.g., Borusyak and Jaravel 2017; Callaway and Sant'Anna 2021; Sun and Abraham 2021; Wooldridge 2021; Borusyak, Jaravel, and Spiess forthcoming; de Chaisemartin and D'Haultfœuille 2020; Wooldridge 2023). The new RTA estimate is still positive and statistically significant. However, it is significantly larger (more than double), implying that the RTAs in our sample have led to an increase of about 46 percent in trade among member countries. Comparisons among the

[2] Calculated as $\left(\exp(0.166) - 1\right) \times 100 = 18.057$. This estimate is comparable to recent results from the literature. For example, Baier, Yotov, and Zylkin (2019) obtain an estimate of the impact of free trade agreements of 0.19, implying a corresponding trade volume effect of about 21 percent. The preferred estimate from the survey of Larch and Yotov (2023) is similar (0.18).

alternative estimates in Figure 1 reveal that, together with the use of domestic trade, the impact of the new estimator on the RTA estimate is among the most significant methodological changes in relative terms.[3]

From a methodological perspective, our contribution is to adapt and "nest" the new (heterogeneity-robust) staggered DiD approach within an empirical panel gravity model,[4] which takes into account the best estimation practices from the trade gravity literature, including the use of the nonlinear Poisson pseudo-maximum likelihood (PPML) estimator (Santos Silva and Tenreyro 2006) and multidimensional fixed effects (Yotov et al. 2016). While there is an array of excellent recent heterogeneity-robust DiD estimators for linear settings,[5] treatments of nonlinear DiD with staggered interventions, such as PPML, are limited. Therefore, to implement our methods, we rely on Wooldridge (2023), who generalizes the linear setting from Wooldridge (2021), which proposes an *extended two-way fixed effect* (*ETWFE*) *estimator*, for the estimation of nonlinear models.[6] The ETWFE estimator maintains the general structure of the basic two-way fixed effect (TWFE) regression, but it also allows for treatment effect heterogeneity by additionally introducing suitable cohort (i.e., all units treated in a particular year) and year interactions.

In terms of the existing literature, our contribution is most closely related to the gravity RTA literature, where the "average treatment effect" methods of Baier and Bergstrand (2007) have become the standard tool to estimate the effects of RTAs. In addition, and more importantly for our purposes, recent trade papers have shown that the average estimates of the effects of trade agreements mask significant heterogeneity both across treatment groups and also across different time periods—for example, Baier, Yotov, and Zylkin (2019) and Larch and Yotov (2023). At the same time, estimation methods similar to the ones commonly used in the gravity setting have recently come under considerable scrutiny from econometric papers that analyze settings with staggered DiD designs, where the estimates are possibly biased in the presence of treatment effect heterogeneity, for example, due to so-called forbidden comparisons that (mis)use already-treated units in the control group (e.g., Borusyak and Jaravel 2017; de Chaisemartin and D'Haultfœuille 2020).

We also see two broader implications of our staggered DiD methods of estimating nonlinear gravity models with high-dimensional fixed effects. First, in relation to the existing trade literature, the proposed methods can be used to revisit and improve the estimates from a series of policy applications that have been of significant interest to trade economists, including the effects of membership in the

[3] Another notable result from our analysis is based on an event study, which we perform in Section IIIC, and it reveals that the new RTA estimates are not only larger but also spread over a longer period of time. A battery of sensitivity experiments, which are based on insights from the recent DiD literature and from the empirical trade literature, confirm the robustness of our main findings.

[4] The gravity equation, for example, in Eaton and Kortum (2002) and Anderson and van Wincoop (2003), is the workhorse model of international trade, and it has been used in tens of thousands of applications.

[5] See, for example, Callaway and Sant'Anna (2021), Sun and Abraham (2021), Wooldridge (2021), Borusyak, Jaravel, and Spiess (forthcoming), and de Chaisemartin and D'Haultfœuille (2020). de Chaisemartin and D'Haultfœuille (2023) and Roth et al. (2023) offer surveys of this literature.

[6] Strictly speaking, it is the estimation *model* rather than the estimation *method* that is extended, that is, the estimation method remains the same but is applied to a more flexible model. We offer a more detailed discussion of the implementation and implications of ETWFE in Section I. We thank Jeff Wooldridge for suggesting this clarification.

World Trade Organization (WTO) (e.g., Rose 2004 and Larch et al. 2019b), currency unions (e.g., Rose 2000 and Larch et al. 2019a), and economic sanctions (e.g., Hufbauer and Oegg 2003 and Felbermayr et al. 2020). Second, we believe that our methods can improve on gravity specifications of bilateral flows beyond trade, including migration (e.g., Anderson 2011), FDI (e.g., Anderson, Larch, and Yotov 2019), commuting patterns (e.g., Persyn and Torfs 2016), and, more generally, to any structural or atheoretical setting of bilateral flows that is subject to economic gravity forces.

The rest of the paper is structured as follows. Section I describes our methods. Section II provides information on our data and the data sources that we use. Section III presents our main findings. Section IV provides an overview of our robustness analyses. Section V concludes. A supplementary online Appendix contains additional descriptive statistics and estimation results from a series of robustness experiments.

I. Methods

The objective of this section is to "nest" the insights from the new heterogeneity-robust staggered DiD methods within a structural gravity model. To this end, in Section IA, we review the current best practices to estimate the gravity model of trade. Then, departing from the benchmark gravity setting, in Section IB, we implement and discuss the implications of the relevant developments in the recent staggered DiD literature. For expositional purposes, and consistent with our empirical analysis, our focus in this section will be on the effects of RTAs.

A. *Estimating Structural Gravity: A Brief Review*

Our departing point is the following econometric panel gravity model, which takes into account the current best practice recommendations from the existing literature:[7]

$$Y_{ij,t} = \exp\left\{\delta^{TWFE} RTA_{ij,t} + \pi_{i,t} + \chi_{j,t} + \tau_{ij} + \theta_{ii,t}\right\} \times \epsilon_{ij,t}. \tag{1}$$

Here, $Y_{ij,t}$ denotes nominal trade flows (Baldwin and Taglioni 2006) from country i to country j at time t, including domestic trade flows (Yotov 2022). To account for heteroskedasticity and to take advantage of the information contained in zero trade flows, we use the PPML estimator (Santos Silva and Tenreyro 2006, 2021). $\pi_{i,t}$ are exporter-year fixed effects, and $\chi_{j,t}$ are importer-year fixed effects, which account for the multilateral resistance terms of Anderson and van Wincoop (2003) as well as for any other country-time determinants of trade flows on the exporter and on the importer side (Hummels 2001; Baldwin and Taglioni 2006; Olivero and Yotov 2012). τ_{ij} are directional pair fixed effects (Baier, Yotov, and Zylkin 2019), which mitigate endogeneity concerns (Baier and Bergstrand 2007) and

[7] For brevity, we only offer motivation and representative references for each of the terms in our gravity specification (1). We refer the reader to Larch and Yotov (2023) for a complete bibliography and a detailed discussion of the evolution of the gravity methods with a focus on RTAs.

control for all symmetric and asymmetric time-invariant trade costs (Egger and Nigai 2016; Agnosteva, Anderson, and Yotov 2019). $\theta_{ii,t}$ denotes a set of dummy variables that are equal to 1 for domestic trade and 0 for international trade for each year in the sample.[8] These covariates account for common globalization trends, such as improvements in communication, transportation, etc., (Bergstrand, Larch, and Yotov 2015). Lastly, $\epsilon_{ij,t}$ is a multiplicative error term, and we cluster all standard errors by country-pair (Egger and Tarlea 2015).[9]

Most importantly for our purposes, δ^{TWFE} is the TWFE estimate of the effect of RTAs on trade, whose estimate is of central interest to us. While, from a structural gravity perspective, the estimate of the effects of RTAs could vary across the same dimensions as the bilateral trade flows data itself—that is, "ij,t"[10]—the vast majority of RTA applications impose a common/average effect of RTAs (Baier and Bergstrand 2007; Anderson and Yotov 2016). Only recently have there been some attempts to allow for certain heterogeneity in the effects of RTAs, for example, across agreements, pairs, or time (Baier, Yotov, and Zylkin 2019; Larch and Yotov 2023). However, as noted earlier, the existing RTA estimates from the gravity literature are possibly biased in the presence of treatment effect heterogeneity, for example, due to so-called forbidden comparisons that (mis)use already-treated units in the control group (Borusyak and Jaravel 2017; de Chaisemartin and D'Haultfœuille 2020), and this is the main motivation for our current contribution and analysis.

B. *Gravity with Heterogeneity-Robust Staggered DiD*

The TWFE estimator has recently come under considerable scrutiny in settings with staggered DiD designs, that is, in which an intervention occurs in multiple units in different time periods. In the presence of heterogeneous treatment effects, that is, if the effect of the intervention is heterogeneous between groups or over time, which is exactly the case with many of the policies that are evaluated with the gravity model, the TWFE estimator may result in biased estimates that are difficult to interpret. As shown by Borusyak and Jaravel (2017) and de Chaisemartin and D'Haultfœuille (2020), TWFE estimates are a weighted sum of average treatment effects on the treated (ATTs) in each group and time period, with weights that may be negative. As a consequence, TWFE regressions may not identify a convex combination of treatment effects; that is, the TWFE coefficient may be negative even if all individual ATTs are positive. The source of the bias lies in so-called forbidden comparisons, in which units that are treated in early periods (already-treated units) are included in the control group for units treated in later periods (Borusyak and Jaravel 2017).

[8] For example, $\theta_{ii,2000}$ is a dummy variable that takes the value of 1 for domestic trade in 2000 and otherwise 0.

[9] In some robustness checks, we also additionally include a vector of gravity controls, such as distance, contiguity, language, colonial ties, WTO membership, GDP, and remoteness indexes. We refer the reader to Yotov et al. (2016) for a discussion of the alternative practices for gravity estimations.

[10] The reason is that, regardless of whether the gravity model is derived on the demand side, for example, in Anderson and van Wincoop (2003) or on the supply side, for example, in Eaton and Kortum (2002), the coefficient on the RTA variable is a function of two parameters—the direct elasticity of trade with respect to RTAs and the trade elasticity. As demonstrated by Carrère, Mrázová, and Neary (2020), the latter can vary across the dimensions of the trade data. Thus, the composite elasticity δ^{TWFE} should vary across the same dimensions. We thank an anonymous referee for asking us to add a discussion about the dimensionality of the RTA estimate from a structural gravity perspective.

For the linear setting, a wide range of heterogeneity-robust DiD estimators have been proposed in the recent literature (e.g., de Chaisemartin and D'Haultfœuille 2020; Callaway and Sant'Anna 2021; Sun and Abraham 2021; Wooldridge 2021; Borusyak, Jaravel, and Spiess forthcoming), which differ inter alia in their identifying assumptions, comparison groups, and efficiency properties.[11] An explicit treatment of nonlinear DiD with staggered interventions, like the gravity setting considered in equation (1), is more limited. For the estimation of nonlinear models, Wooldridge (2023) proposes an extended TWFE estimator analogous to the linear case considered in Wooldridge (2021). The extended TWFE estimator maintains the general structure of the basic TWFE regression but allows for treatment effect heterogeneity at the cohort-year level by additionally introducing suitable cohort and year interactions.[12]

We preserve all recommendations from the gravity literature, and, following Wooldridge (2023), we make one important adjustment to equation (1). Specifically, we replace the δ^{TWFE} coefficient and the single indicator variable for the presence of an RTA between i and j at time t ($RTA_{ij,t}$) with the following term:

$$\sum_{g=q}^{T} \sum_{s=g}^{T} \delta_{gs} D_{gs}, \tag{2}$$

where country pair ij belongs to treatment cohort g if the RTA onset was in year g, q is the first year of the treatment of cohort g, T is the last year of the panel, D_{gs} is a time-varying treatment indicator equal to 1 for cohort g for $s{=}t$ in posttreatment years and 0 otherwise, and δ_{gs} captures the cohort-year-specific treatment effects.[13] Equation (2) avoids comparisons with already-treated units by saturating the respective observations with fixed effects.[14]

Taking into account the above considerations, our main estimating equation becomes

$$Y_{ij,t} = \exp\left\{ \sum_{g=q}^{T} \sum_{s=g}^{T} \delta_{gs} D_{gs} + \pi_{i,t} + \chi_{j,t} + \tau_{i,j} + \theta_{ii,t} \right\} \times \epsilon_{ij,t}. \tag{3}$$

As noted earlier, the only difference between equations (3) and (1) is our treatment of RTAs. In the empirical analysis, we will obtain and compare the corresponding

[11] See, for example, de Chaisemartin and D'Haultfœuille (2023) and Roth et al. (2023) for recent surveys of the literature. See also, for example, Cengiz et al. (2019), Deshpande and Li (2019), Egger, Koethenbuerger, and Loumeau (2022), and Gardner (2022) for stacked regression approaches.

[12] Note that, like many other estimators in the literature (e.g., de Chaisemartin and D'Haultfœuille 2020; Callaway and Sant'Anna 2021; Sun and Abraham 2021), the extended TWFE with cohort-year interactions in principle allows for arbitrary heterogeneity within cohort-year cells but can only identify (simple) average treatment effects at the cohort-year level. If one is interested in different (weighted) averages of cohort-year effects or even in individual treatment effects, additional interactions or an imputation approach are necessary (Borusyak, Jaravel, and Spiess forthcoming). We offer further related discussion in Section IV, where we also relax these restrictions by considering an ETWFE specification, which allows for heterogeneity at the agreement-year level, as well as a Poisson imputation estimator (Wooldridge 2023).

[13] For example, $\delta_{1990,1990}$ captures the treatment effect of all RTAs signed in the year 1990 on trade in the first year, and $\delta_{1990,1991}$ captures the treatment effect of the same RTAs on trade in the second year, etc.

[14] Note that in the current form, equation (2) includes both not-yet-treated and never-treated units as controls (e.g., Callaway and Sant'Anna 2021), though the comparison group can also be restricted (see also Section IV).

estimates from the TWFE and ETWFE specifications.[15] Before that, we explicitly state under which conditions the treatment effects of interest are identified, discuss the aggregation of the cohort-year-specific treatment effects, and illustrate the potential bias of the TWFE with an example.

Identifying Assumption.—Initially, we are interested in identifying cohort-time-specific (approximate) proportional treatment effects. For brevity, in the following, we will often refer to these simply as "treatment effects." Inference on ATTs, that is, the level effects, is somewhat complicated in this setting by the need to obtain consistent coefficient estimates including standard errors for all fixed effects in equation (3) (Wooldridge 2023). Following Wooldridge (2023) in the potential outcome notation of Athey and Imbens (2022), the proportional treatment effects are for all $g \in \{q, \ldots, T\}$ and $s \in \{g, \ldots, T\}$

$$\delta_{gs} = \ln\left(\frac{E\left[Y_s(g)|D_g = 1\right]}{E\left[Y_s(\infty)|D_g = 1\right]}\right) \tag{4}$$

$$= \ln\left(E\left[Y_s(g)|D_g = 1\right]\right) - \ln\left(E\left[Y_s(\infty)|D_g = 1\right]\right),$$

where $Y_t(g)$ is the potential outcome of cohort g at time t, $Y_t(\infty)$ is the potential outcome in the never treated state—that is, of units not subjected to the treatment over a particular time period—and D_g is a binary variable indicating membership to cohort g. To identify δ_{gs}, two assumptions are necessary, which we explicitly test in Section IIIA.[16]

First, the no-anticipation assumption requires for all $g \in \{q, \ldots, T\}$ and $t \in \{1, \ldots, g-1\}$ that

$$E\left[Y_t(g)|D_g = 1, \mathbf{X}\right] = E\left[Y_t(\infty)|D_g = 1, \mathbf{X}\right], \tag{5}$$

where $\mathbf{X}$ is a vector of covariates subsuming all exporter-year, importer-year, and border-year fixed effects. In words, the potential outcomes prior to treatment are, on average and conditional on the fixed effects, the same—that is, there is no effect of the treatment before its onset. As discussed, for example, in Callaway and Sant'Anna (2021) and Wooldridge (2023), in principle, the no-anticipation assumption does not need to hold in all pretreatment periods and can be relaxed by omitting conspicuous periods at the cost of efficiency.

Second, the parallel trends assumption requires for all $t \in \{1, \ldots, T\}$ that

$$\frac{E\left[Y_t(\infty)|D_q, \ldots, D_T\right]}{E\left[Y_1(\infty)|D_q, \ldots, D_T\right]} = \exp\{\pi_t\}, \tag{6}$$

[15] In terms of practical implementation, we augmented a previous version of the Stata command *jwdid* (Rios-Avila 2022) to account for the best estimation practices from the gravity literature, which we summarized in Section IA. The current version of *jwdid* now allows for estimation with ETWFE in the gravity setting.

[16] Note that equation (3) differs from the setting considered in Wooldridge (2023) with regard to its additional fixed effect structure (against the background of the more complex double-indexed panel data under consideration).

where π_t captures all the coefficients corresponding to exporter-year, importer-year, and border-year fixed effects. In words, the parallel trends assumption states that the growth in the outcome of the treatment and control group would have been the same in the absence of treatment; that is, growth may depend on the time-varying covariates $\mathbf{X}$ but not on the treatment $\mathbf{D}_i = \left(D_{iq}, \ldots, D_{iT}\right)$. Note that due to the use of the exponential mean in equation (3), the parallel trends assumption is stated in terms of the ratio of means rather than the difference in means as in the linear case (see also, e.g., Athey and Imbens 2006; Ciani and Fisher 2019).

Lastly, two additional points are worth highlighting. First, just like the TWFE estimator, the ETWFE identifies partial effects of RTAs on trade due to the inclusion of exporter-year and importer-year fixed effects, which capture multilateral resistance terms and are expected to change as a result of RTAs. The total effect of RTAs on trade, including these general equilibrium effects could, in principle, be backed out using computational approaches that are now standard in the trade literature and can even be performed with the PPML estimator directly in Stata, for example, as described in Anderson, Larch, and Yotov (2018). Second, note that there is a certain asymmetry between the restrictions on nontreated potential outcomes and restrictions on the treatment effects (see footnote 10 and page 8 of Borusyak, Jaravel, and Spiess forthcoming). Here, we follow the literature on heterogeneity-robust DiD estimators, which takes the view that restrictions on the form of nontreated potential outcomes are acceptable, while homogeneity restrictions on treatment effects are not, which is not a limitation that is specific to our paper.

Estimation Target.—The ETWFE estimator proposed in Wooldridge (2021, 2023) initially requires estimation of treatment effects at a very granular level, which is a common feature in many heterogeneity-robust DiD estimators proposed in the recent literature (e.g., Callaway and Sant'Anna 2021; Sun and Abraham 2021; Borusyak Jaravel and Spiess forthcoming; de Chaisemartin and D'Haultfœuille 2020). However, interest often lies in more aggregated quantities rather than the directly estimated cohort-time-specific treatment effects from equation (3). Consistent with the large corresponding trade literature, in this paper, we are interested in evaluating how much international trade has grown, on average, in all country pairs and postintervention years as a result of the introduction of RTAs. Therefore, our main estimation target is a single RTA estimate, which is defined by the weighted sum of the estimated cohort-time-specific treatment effects as

$$\bar{\hat{\delta}} = \sum_{g=q}^{T} \sum_{s=g}^{T} \frac{N_{gs}}{N_D} \hat{\delta}_{gs}, \tag{7}$$

where we assign equal weight to all posttreatment observations, which correspond to the number of observations of cohort g in period s, N_{gs}, relative to the total number of treated observations, $N_D = \sum_{g=q}^{T} \sum_{s=g}^{T} N_{gs}$. Similarly, the standard errors for the ETWFE estimator are computed as a (weighted) linear combination of the cohort-year-specific effects taking the covariance between the coefficients into account. The reasons for the selection of these weights are that we view them as the simplest, most intuitive, and most transparent option, which is also consistent with de Chaisemartin and D'Haultfœuille (2020) and Borusyak, Jaravel, and Spiess (forthcoming).[17] Naturally, in other settings and for other research questions, different weighting schemes may be more appropriate.[18]

Note that, since $\hat{\delta}_{gs}$ is defined by a log difference (cf. equation (4)), equation (7) has the attractive property that it is directly related to the geometric mean, as defined by the arithmetic mean in logscale. Therefore, $\exp\left(\hat{\delta}\right) - 1$ computes the geometric growth rate of the RTA effect on trade, which takes account of compounding and is more appropriate than the arithmetic mean in our setting with exponential growth.

We also consider two additional estimation targets. First, we consider a cohort estimand, for which we compute cohort-specific treatment effects by averaging over the time dimension as follows:

$$\overline{\hat{\delta}_{g\cdot}} = \sum_{s=g}^{T} \frac{N_{gs}}{N_{g\cdot}} \hat{\delta}_{gs}, \tag{8}$$

where $N_{g\cdot} = \sum_{s=g}^{T} N_{gs}$ is the total number of posttreatment observations of cohort g. Second, we consider an event-study estimand, for which we compute event-time-specific treatment effects by averaging over the cohort dimension as follows:

$$\overline{\hat{\delta}_{\cdot s}} = \sum_{g=q}^{s} \frac{N_{gs}}{N_{\cdot s}} \hat{\delta}_{gs}, \tag{9}$$

where $N_{\cdot s} = \sum_{g=q}^{s} N_{gs}$ is the total number of treated observations in period s. Event-time-specific treatment effects are akin to results from a dynamic TWFE (or event-study) specification without its shortcomings based on the heterogeneity-robust innovations in the DiD literature (e.g., Sun and Abraham 2021; Borusyak, Jaravel, and Spiess forthcoming), while cohort-specific treatment effects are usually not studied in the associated TWFE literature.

As discussed in detail in Callaway and Sant'Anna (2021), caution needs to be exercised when comparing $\overline{\hat{\delta}_{g\cdot}}$ across cohorts or $\overline{\hat{\delta}_{\cdot s}}$ across time periods since compositional changes due to differences in weights and in the composition of groups/periods can complicate their interpretation. For example, early-treated cohorts have more observations many years after the treatment onset than late-treated cohorts. Consequently, in the presence of treatment effect heterogeneity across cohorts, changes in event-time-specific treatment effects are the result of changes in event time as well as the higher share of early-treated cohorts relative to late-treated cohorts. Similarly, in the presence of treatment effect heterogeneity across event time, differences in cohort-specific treatment effects are due to differences across cohorts as well as the higher share of long-term effects in early-treated cohorts relative to late-treated cohorts.

[17] In the supplementary online Appendix, we also consider the robustness of our results to alternative weighting schemes.

[18] See also the discussion on this point in Borusyak, Jaravel, and Spiess (forthcoming).

On the Bias of the TWFE Estimator: An Illustrating Example.—This subsection provides a small stylized example to illustrate the potential bias of the TWFE estimator in the presence of treatment effect heterogeneity. Here, we focus on the OLS TWFE estimator, which can be shown to identify some weighted average of the true underlying heterogeneous cohort-year effects under standard identifying assumptions (Borusyak and Jaravel 2017; de Chaisemartin and D'Haultfœuille 2020), and we consider the example studied in de Chaisemartin and D'Haultfœuille (2020) and Borusyak, Jaravel, and Spiess (forthcoming) with two equal-sized cohorts (or units) and three time periods.

The early-treated cohort A is treated in period 2, and the late-treated cohort B is treated in period 3. The treatment effect for cohort A in periods 2 and 3 is denoted by δ_{A2} and δ_{A3}, respectively, and the treatment effect for cohort B in period 3 is denoted by δ_{B3}. The outcome of cohort i in period t is denoted by Y_{it}. The TWFE estimand can be shown to equal $\delta^{TWFE} = \delta_{A2} - \frac{1}{2}\delta_{A3} + \frac{1}{2}\delta_{B3}$ under a parallel trends and a no-anticipation assumption (Borusyak, Jaravel, and Spiess forthcoming). By contrast, the corresponding ETWFE estimand from Section IB would be $\delta^{ETWFE} = \frac{1}{3}\delta_{A2} + \frac{1}{3}\delta_{A3} + \frac{1}{3}\delta_{B3}$, that is, the weight equals the proportion of each cohort-year cell in all posttreatment observations.

This example illustrates several points. First, note that if the assumption of treatment effect homogeneity of the TWFE estimator is fulfilled, then the TWFE estimand correctly identifies the true underlying treatment effect ($\delta^{TWFE} = \delta_{A2} = \delta_{A3} = \delta_{B3}$). Second, the problem of the TWFE estimator in the presence of treatment effect heterogeneity results from "forbidden comparisons" (Borusyak and Jaravel 2017). An "admissible comparison" would be to compare the evolution of cohort A and B between periods 2 to 1, $(Y_{A2} - Y_{A1}) - (Y_{B2} - Y_{B1})$. By contrast, in this example, the TWFE estimator also uses a "forbidden comparison," in which the evolution of cohort B and A between periods 3 to 2, $(Y_{B3} - Y_{B2}) - (Y_{A3} - Y_{A2})$, is used. However, subtracting the term $(Y_{A3} - Y_{A2})$, inter alia, also subtracts the evolution of treatment effects, $\delta_{A3} - \delta_{A2}$, thereby placing a negative weight on the late treatment effect of the early-treated cohort A (Borusyak, Jaravel, and Spiess forthcoming).

Third, in comparison to the ETWFE estimand, the TWFE estimand underweights early-treated cohorts and overweights late-treated cohorts. For example, in the case considered above, the weight placed on cohort A (B) is $\frac{1}{2}$ ($\frac{1}{2}$) for the TWFE estimand and $\frac{2}{3}$ ($\frac{1}{3}$) for the ETWFE estimand. If treatment effects decrease (increase) over cohorts, this results in a downward (upward) bias of the TWFE estimand. As demonstrated in the empirical analysis below, this is exactly the case for the effects of RTAs, thus highlighting the benefits of the ETWFE estimator.

Fourth, in comparison to the ETWFE estimand, the TWFE estimand overweights short-run effects and underweights long-run effects resulting in a severe short-run bias of the static TWFE specification (Borusyak, Jaravel, and Spiess forthcoming). For example, in the case considered above, the weight placed on the first (second) year of treatment is $1\frac{1}{2}$ ($-\frac{1}{2}$) for the TWFE estimand and $\frac{2}{3}$ ($\frac{1}{3}$) for the ETWFE estimand. If treatment effects increase (decrease) over time, this also results in a downward (upward) bias of the TWFE estimand. Like above, we also find empirical support for this phenomenon for the effects of RTAs, which are typically characterized by strong maturation effects.

Fifth, de Chaisemartin and D'Haultfœuille (2020) generalize the results in points three and four. They show that negative weights are more likely to occur in time periods when a large fraction of groups are treated and in groups treated for many periods (i.e., late treatment effect of the early-treated cohorts).

Lastly, note that the extent of negative weights depends on a number of factors and that they might entirely disappear with a large number of never-treated and not-yet-treated observations (Borusyak, Jaravel, and Spiess forthcoming). However, even in this case, the weights of the TWFE estimand might strongly diverge from those of the ETWFE estimand in the direction described above.[19]

II. Data

Given the methodological purpose of our paper, for our empirical analysis, we rely on three standard datasets, including data on (i) trade flows, (ii) RTAs, and (iii) other gravity variables. The trade data that we use to obtain our estimates are from the Structural Gravity Database (SGD) of the WTO (Larch et al. 2019b; WTO 2019).[20] The SGD contains aggregate manufacturing trade data and has three important advantages for our purposes. First, it covers a long period of time, 1980–2016. Second, it contains many countries (a total of 229). In combination, this implies that we could have "long T" and "long N" panel datasets for our analysis, and we experiment with alternative sample sizes.[21] Third, the SGD includes consistently constructed domestic and international trade flows, and we demonstrate that our methods have implications for gravity estimations with and without domestic trade flows.[22]

In addition to the SGD, we use the Dynamic Gravity Dataset (DGD) of the United States International Trade Commission (Gurevich and Herman 2018; United States International Trade Commission 2021a, b) for data on trade agreements and some other standard gravity variables (e.g., bilateral distance, contiguity, WTO membership, etc.) that we employ in the robustness analysis.[23] This database covers all trade agreements in the world that entered into force during the period for which trade data are available.[24] As briefly discussed in the introduction, the focus on RTAs is suitable for our purposes for several reasons. First, the impact of RTAs is probably the most studied topic from a trade policy perspective. In addition, there is ample empirical evidence that the effects of RTAs are heterogeneous across groups, such as

[19] We offer an analysis of the underlying weights of the OLS TWFE estimator in our setting in Section IIID.

[20] The SGD is available at https://www.wto.org/english/res_e/reser_e/structural_gravity_e.htm.

[21] To obtain our main results, we use a sample of 69 countries, which covers 98 percent of world exports. We also experiment with a medium sample, which contains 91 countries covering 99 percent of world exports, and also with a large sample, which covers all the countries from the SGD. Due to singletons, the actual number of countries in our large estimating sample is 225 (out of 229).

[22] The use of domestic trade flows, in addition to international trade, is consistent with gravity theory and has a number of potential advantages for gravity estimations. Yotov (2022) reviews the literature.

[23] The DGD is freely available at https://www.usitc.gov/data/gravity/dgd.htm.

[24] Other excellent RTA datasets include, for example, the Regional Trade Agreements Database of Egger and Larch (2008) and the NSF-Kellogg Institute Data Base on Economic Integration Agreements of Baier and Bergstrand (2021).

between agreements and pairs within agreements (Baier, Yotov, and Zylkin 2019). Thus, existing RTA estimates may indeed be subject to the recent critiques from the staggered DiD literature.

A known challenge with the existing RTA datasets is that they report the date of the effective entry into force of the RTA as the initial RTA date. However, in reality, the onset of the effects of RTAs usually does not coincide with the date of their entry into force. The reason is that once an RTA is announced, which usually happens several years prior to its effective implementation, some trade barriers may start falling, and/or some economic agents may start adjusting in anticipation of the RTA (e.g., Moser and Rose 2012a, 2014; Egger, Larch, and Yotov 2022). Thus, arguably, the announcement of the RTA negotiations is probably the more appropriate onset for studying the impact of RTAs.

Unfortunately, as noted earlier, we are not aware of a comprehensive dataset on the announcement of RTAs. However, there is some evidence that could guide us. For example, according to Moser and Rose (2012b), the period between the start of the initial negotiations and the signing of a trade agreement is about 28 months. In addition, according to the WTO data, the average period between the signing and the entry into force of an agreement is about one year. In combination, this implies that, on average, the RTA negotiations start about three years before the date of entry into force. This is consistent with the results from Egger, Larch, and Yotov (2022), who find that the impact of RTAs begins about three years prior to their entry into force, possibly at the time when they are announced or signed.

Based on this, for our main analysis, we will assume an "onset" of RTAs three years before their entry into force.[25] As a robustness test, following Wooldridge (2023), we also omit these time periods in Section IV. For econometric reasons, we exclude country pairs with an RTA onset prior to 1980 (so-called always-treated units) since the RTA effect on trade cannot be identified in these cases given the absence of pretreatment periods (and since they also cannot serve as control observations in the presence of treatment effect heterogeneity).

Table A1 in the supplementary online Appendix reports the number of observations, country pairs, exporters, importers, and years for different groups in the dataset. A cohort refers to all country pairs with an RTA onset in a particular year. First, online Appendix Table A1 highlights that the dataset contains a reasonable number of pre- and posttreatment observations for most cohorts. Second, online Appendix Table A1 captures the sharp increase in RTAs in the late 1980s and 1990s. Another feature of the evolution of the RTAs, which is not obvious from online Appendix Table A1, is that trade agreements have become increasingly "deep" over time—that is, they contain more provisions for trade liberalization. This type of heterogeneity could further motivate the need for the proposed ETWFE estimator. Third, to state the obvious, early-treated cohorts have more posttreatment years than late-treated cohorts, implying that longer-run effects can only be estimated for the former. Fourth, note that the dataset contains two control groups, never-treated country pairs and not-yet-treated country pairs, that can be used for comparisons

[25] Note that we only consider RTAs that entered into force in the time period up until 2016 in our sample.

with the posttreatment years of the treatment group. Never-treated country pairs are those that did not sign an agreement in the time period of our sample. Not-yet-treated country pairs are those without an RTA onset up until the year of the comparison but with an RTA onset in later years of the sample.

Lastly, Table A2 in the supplementary online Appendix also reports summary statistics of standard gravity variables for different groups in the dataset. We note that distance, in particular, varies strongly across different treatment cohorts. In Section IV, we exploit this variation in an additional specification akin to a regression adjustment approach (Heckman, Ichimura, and Todd 1997; Wooldridge 2023) that relaxes the parallel trend assumption of the ETWFE estimator.

III. Main Results and Analysis

This section presents our main findings. We develop the analysis in four steps. First, we start by testing the identifying assumptions of the ETWFE estimator (Section IIIA). Second, we obtain and discuss our main results (Section IIIB). Third, we present event-study-type estimates that describe the evolution of the RTA effects over time, as well as a series of cohort-specific RTA estimates (Section IIIC). Fourth, we compute the implicit weights underlying the (OLS) TWFE estimator (Section IIID).

A. *Test of Identifying Assumptions*

The ETWFE estimator requires the parallel trends assumptions and the no-anticipation assumption (described in detail in Section IB) to hold. Therefore, we begin by testing these two assumptions. To this end, we adapt to our setting the methods of Borusyak, Jaravel, and Spiess (forthcoming).[26] Specifically, we replace the treatment effects in equation (3) with cohort-time-specific placebo treatment effects prior to treatment onset and estimate the regression with untreated observations only. To avoid misleading visual inferences, we follow the recommendation by Roth (2024) and put our pretreatment estimates in a different plot than the posttreatment estimates. Figure 2 provides evidence against the violation of the identifying assumptions by showing the aggregated placebo effects in event time relative to treatment onset. The pretreatment coefficients are all close to zero, and a joint hypothesis test is insignificant (p-value of 0.2978). Therefore, we conclude that there are no signs of significant pre-trends in our setting, and we proceed to obtain our main results. While this is not a limitation specific to this paper, note that, in particular for early cohorts, we can only test the parallel trends assumption over a short time period, whereas that assumption has to hold for much longer time periods for all of our event-study estimates to be unbiased.

[26] Alternatively, using the parallel trend test based on the entire sample discussed in Wooldridge (2023) yields similar results. We note that in this setting the two tests are not equivalent due to the more complex fixed effect structure under consideration.

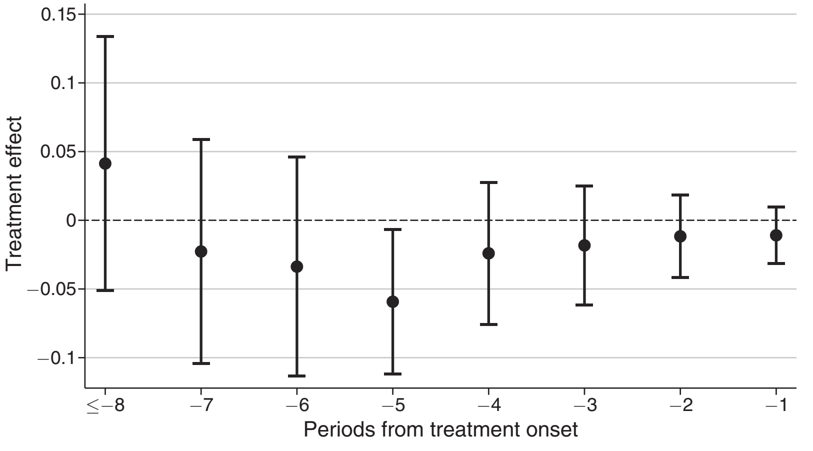

Figure 2. Pretreatment Effects

Notes: The figure reports pre-trend estimates from a PPML estimation of equation (3), in which treatment effects were replaced by cohort-time-specific placebo treatment effects prior to treatment onset. The regression is estimated with untreated observations only in the spirit of Borusyak, Jaravel, and Spiess (forthcoming). The cohort-time-specific treatment effects were aggregated using equation (9) to obtain event-time-specific treatment effect estimates. The figure shows 95 percent confidence intervals using standard errors clustered by country pair.

B. *Main Estimation Results*

Our main estimates are reported in Table 1. The RTA estimate in column 1 corresponds to the "Consecutive years: 2022" result from Figure 1 and is obtained with all current techniques from the gravity literature, including the PPML estimator, exporter-time and importer-time fixed effects, pair fixed effects, domestic trade flows, globalization effects, consecutive-year data, and pair-clustered standard errors as specified in equation (1). The RTA estimate is positive and statistically significant (0.166, standard error 0.050), implying an increase in the bilateral trade between member countries of about 18 percent (calculated as $\left(\exp(0.166) - 1\right) \times 100 = 18.057$, standard error 5.913), where the standard error is obtained with the delta method.

Our ETWFE RTA estimate appears in column (2) of Table 1, and it is obtained from equation (3), that is, the same specification from column (1), the only difference being that, in addition to all gravity estimation techniques, we also implement a heterogeneity-robust staggered DiD method. Note that the cohort-time-specific treatment effects were aggregated using equation (7) to obtain the aggregate treatment effect estimate. The aggregate ETWFE estimate is positive and statistically significant, as expected.[27] However, economically, it is substantially larger than the corresponding TWFE result from column (1), implying that the RTAs in our sample have led to an increase of about 46 percent in trade among member countries. A

[27] See Section IV for a discussion of the standard error estimate.

Table 1—Comparison of TWFE and ETWFE Estimators for Different Samples

	TWFE (1)	ETWFE (2)	TWFE (3)	ETWFE (4)	TWFE (5)	ETWFE (6)
$RTA_{ij,t}$	0.166	0.381	0.167	0.327	0.165	0.293
	(0.050)	(0.041)	(0.048)	(0.040)	(0.047)	(0.039)
Sample	Baseline	Baseline	Medium	Medium	Large	Large
Observations	105,409	105,409	175,796	175,796	591,092	591,092
Exporters	69	69	91	91	225	225
Importers	69	69	91	91	225	225
Years	34	34	34	34	34	34
Coefficients	1	469	1	469	1	528
p-value H0: TWFE $=$ ETWFE		0.000		0.000		0.001
$RTA_{ij,t}$ (jackknife)	0.200	0.417	0.200	0.365	0.199	0.326
	(0.067)	(0.090)	(0.060)	(0.075)	(0.078)	(0.064)
Exporter $\times$ *importer* FE	Yes	Yes	Yes	Yes	Yes	Yes
Exporter $\times$ *year* FE	Yes	Yes	Yes	Yes	Yes	Yes
Importer $\times$ *year* FE	Yes	Yes	Yes	Yes	Yes	Yes
Cross-border $\times$ *year* FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table presents PPML regression results using the TWFE estimator (equation (1)) and the ETWFE estimator (equation (3)), for which the cohort-time-specific treatment effects were aggregated using equation (7) to obtain an aggregate treatment effect estimate. The dependent variable is exports, which vary over the exporter-importer-year dimension. The "Baseline" sample contains 69 countries, accounting for 98 percent of world exports. The "Medium" sample, which is used to obtain the results in columns 3 and 4, contains 91 countries, accounting for 99 percent of world exports. The "Large" sample, which is used to obtain the results in columns 5 and 6, contains the full set of countries from the structural gravity dataset. "Coefficients" reports the number of estimated coefficients apart from the fixed effects. Standard errors in parentheses are clustered by country pair. The row labeled "p-value H0: TWFE $=$ ETWFE" reports the results of a standard F-test of the equality of the corresponding TWFE and ETWFE coefficients, where the covariance between the two coefficients is accounted for using a seemingly unrelated regression specification. Jackknife estimates are computed according to the procedure described in Section IV and Section A.3.2 in the supplementary online Appendix.

standard F-test confirms that the two coefficients are also statistically significantly different from each other.

Our main result has two implications. First, from a methodological perspective, it reveals that there is still scope for meaningful improvements in the techniques that are used to identify the impact of RTAs within gravity models. Second, the fact that the ETWFE estimate is significantly larger is important from a policy perspective since, as discussed earlier, some of the existing RTA estimates are viewed as unrealistically small.

The rest of the results in Table 1 are obtained with two alternative samples. Specifically, the estimates in columns (3) and (4) are based on a sample that contains 91 countries, accounting for 99 percent of world exports, while the estimates in columns (5) and (6) are obtained with the full set of countries from the WTO dataset. We draw the following conclusions based on comparisons between the estimates that we obtain with the alternative samples. The TWFE estimate remains very stable, while the ETWFE estimate decreases with the increase of the number of countries in our sample. As a result, the gap between the TWFE and ETWFE estimates becomes smaller; however, the ETWFE estimate is still about twice as large as the corresponding TWFE estimate.

To investigate the source of the difference in the ETWFE estimates across the different samples, we compute separate treatment effects for the treated country pairs in the original sample and those that are additionally included in the medium and large sample. The treatment effect for the original treatment group remains virtually unchanged (medium sample: 0.376, standard error 0.039; large sample: 0.387, standard error 0.038), while the treatment effect of the new country pairs is substantially smaller (medium sample: 0.236, standard error 0.046; large sample: 0.244, standard error 0.041). This suggests that the differences are driven by smaller treatment effects in the additional country pairs rather than changes in the composition of the control group. From a methodological perspective, the TWFE estimator might miss these changes for the following reason. While a larger share of smaller treatment effects are expected to reduce the TWFE estimate, the use of these already-treated country pairs with smaller treatment effects in the control group is also expected to raise the overall RTA estimate. Taken together, these two effects appear to offset each other such that the TWFE estimator misses the lower RTA effects in the larger samples, which underscores the importance of using the ETWFE estimator.

Finally, to mitigate/address potential incidental parameter problems (IPPs), which may arise in nonlinear models with fixed effects (Fernández-Val and Weidner 2016; Jochmans 2017; Pfaffermayr 2021; Weidner and Zylkin 2021), we also compute jackknife estimates of the TWFE and the ETWFE model. See Section IV and Section A.3.2 in the supplementary online Appendix for details. Our results appear in the bottom panel of Table 1. The jackknife estimates confirm our main finding that the ETWFE RTA estimate is significantly larger than the corresponding TWFE estimate. In addition, the jackknife results also suggest that the coefficient estimates of TWFE and ETWFE might both be slightly downward biased, and also that the standard error estimate of the ETWFE seems to be downward biased more than the standard error estimate of the TWFE (except for the large sample). Based on this analysis, we recommend computing jackknife coefficient and standard error estimates for the ETWFE in the three-way gravity setting at least as a robustness check.

C. *Disaggregated Results*

We complement the average RTA estimates from Table 1 with a series of disaggregated results. We start with an event-type analysis of the evolution of the RTA effects over time. Our findings are presented in Figure 3, panel A, where we report standard event-study (dynamic TWFE) results in light color and the corresponding ETWFE results in dark color, which are computed from the cohort-time-specific treatment effects by averaging over the cohort dimension (see Section IB).[28]

Two main results stand out from Figure 3, panel A. First, the TWFE estimates are smaller in each period. This is similar to the smaller average RTA effects from the static TWFE estimator from Table 1 and, therefore, reinforces our main result. Second, according to our dynamic TWFE estimates, the impact of RTAs is exhausted relatively fast, for example, about six years after their initial effects

[28] The event-study (dynamic TWFE) estimator corresponds to a restricted version of the ETWFE estimator, in which treatment effect homogeneity across cohorts is imposed (cf. column 1 in online Appendix Table A3).

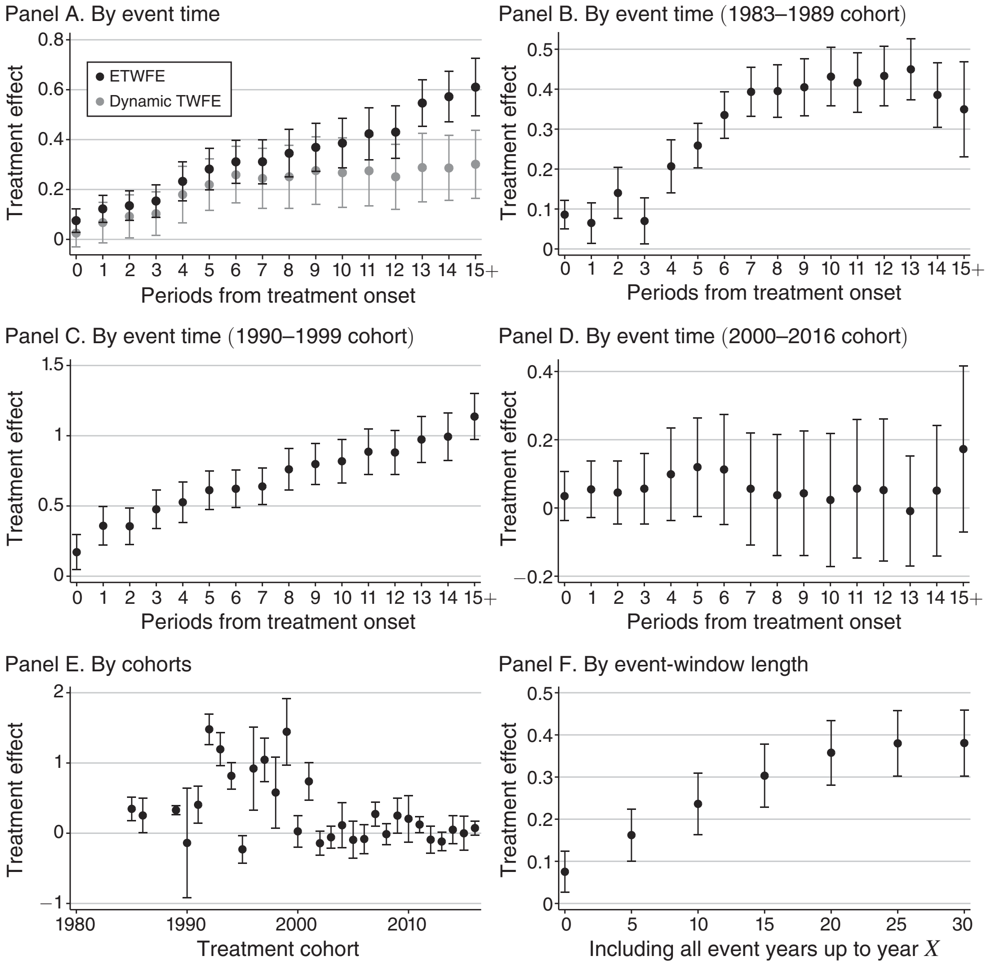

Figure 3. Event-Time-Specific and Cohort-Specific Treatment Effects

Notes: The figure reports different aggregations of cohort-time-specific treatment effects from a PPML estimation of equation (3). Panel A reports event-time-specific treatment effects from equation (3) aggregated using equation (9) in dark color ("ETWFE") along with standard event-study results in light color ("Dynamic TWFE"). Panels B–D report event-time-specific treatment effects from equation (3) aggregated using equation (9) for the 1983–1989 cohort, the 1990–1999 cohort, and the 2000–2016 cohort, respectively. Panel E reports cohort-specific treatment effects from equation (3) aggregated using equation (8). Panel F reports treatment effects aggregated using equation (7), in which only those cohort-time-specific treatment effects are used in the aggregation that are not more than a certain number of years away from treatment onset. The figure shows 95 percent confidence intervals using standard errors clustered by country pair.

are detected.[29] The ETWFE estimates, on the other hand, appear to keep growing slowly but monotonically more than 15 years after the initial impact of the RTAs. This is likely another reason for the difference between the average estimates from columns 1 and 2 of Table 1. In addition, from a policy perspective, the estimates

[29] Note that changes in the estimate over time periods are also affected by changes in the composition of the underlying cohorts since, for example, RTAs signed in later years of the sample are observed for fewer years than RTAs signed in earlier years of the sample.

from Figure 3, panel A imply that the effects of the RTAs in our sample have lasted significantly longer than suggested by the dynamic TWFE estimates.

Next, we capitalize on the advantages of the ETWFE estimator to obtain additional estimates of the effects of RTAs across different dimensions. Figure 3, panels B–D report event-study-type results for three different groups of cohorts—the 1983–1989 cohort, the 1990–1999 cohort, and the 2000–2016 cohort—where, for example, the 1983–1989 cohort refers to all country pairs with an RTA onset between 1983 and 1989. They show that there is substantial heterogeneity across cohorts. For the 1983–1989 cohort, the RTA effect increases and then saturates around ten time periods after treatment onset, while for the 1990–1999 cohort, the RTA effect seems to steadily increase even after 15 years. By contrast, there is no significant effect in any time period for the 2000–2016 cohort.

Figure 3, panel E zooms further in on cohort heterogeneity. It shows average cohort-specific treatment effects, which are computed from the cohort-time-specific treatment effects by averaging over the time dimension (see Section IB). In line with the previous results, there is substantial heterogeneity across treatment cohorts. RTAs with an onset in the 1980s have positive effects—but mostly lower than the average RTA estimates from column 2 in Table 1. The effects of RTAs with an onset in the 1990s are largest and mostly above average, while the cohorts after 2000, on average, do not show significant effects.

We have several intuitive explanations for the heterogeneity of our cohort-specific RTA estimates. First, the large estimates for the cohort of agreements in the 1990s are consistent with the perception of this period as the "golden age" of trade liberalization. Second, on a related note, the most "natural" RTAs were already signed during the 1980s and 1990s, and, therefore, the potential of the more recent agreements, for example, those in the 2000s, to lead to significant increases in trade among members was lower and more limited. Third, the first two decades of the 2000s were marked by significant economic crises (e.g., the dot.com bust and the financial crises), followed by Brexit and the Trump presidency, which all have impacted trade liberalization efforts negatively. Fourth, average applied most-favored-nation tariffs fell significantly over the sample period (e.g., Teti 2020), thereby leaving less scope for trade-enhancing effects of RTA-related tariff reductions.[30]

While a detailed analysis of the underlying determinants of the heterogeneity of RTAs over time is beyond the scope of the current paper, in Section A.3.4 of the supplementary online Appendix, we perform a short ex post subgroup analysis of the heterogeneity of the treatment effects in the spirit of de Chaisemartin and D'Haultfœuille (2024), Shahn (2023), and Baier, Yotov, and Zylkin (2019). Our partial results reveal that bilateral distance, common language, and colonial relationships are important determinants of the effects of RTAs, while the effects of size are not statistically significant. Importantly, from a policy perspective, this analysis

[30] We also recognize that RTAs have become deeper and more comprehensive over time. This may have countervailing effects with respect to the impact of RTAs. On the one hand, deeper trade liberalization should lead to more trade among members. On the other hand, more cumbersome and complicated RTA provisions may impede trade. We explore these hypotheses further in the supplementary online Appendix.

also enabled us to construct a counterfactual scenario, in which all country pairs without an RTA were to sign one. The counterfactual results provide tentative evidence that the RTAs between remaining country pairs would significantly boost trade—in contrast to the results for late-treated cohorts—albeit to a lesser extent than the historical average in the baseline sample.

Lastly, we look at the role that the event-window length and hence years far away from treatment onset play for the magnitude of the aggregate effect. Figure 3, panel F reports average treatment effects in which only those cohort-time-specific treatment effects are used in the aggregation that are not more than a certain number of years away from the treatment onset. In our sample, the aggregate treatment effect steadily increases with event-window length at least up until 20 years and then levels off. This suggests that truncating the sample may bias the aggregate treatment effect estimate downward given the long-lasting dynamic effects of RTAs (e.g., Schmidheiny and Siegloch 2023; Baker, Larcker, and Wang 2022).

We draw two methodological and two policy conclusions based on the analysis in this section. From a methodological perspective, our analysis demonstrates that the ETWFE RTA effects are (i) significantly larger than previously thought based on TWFE estimates and (ii) that the RTA effects last significantly longer than suggested by the dynamic TWFE estimates. From a policy perspective, our analysis reveals that (i) the effects of RTAs phase in gradually and (ii) that recent trade agreements have not been very successful in promoting international trade.

D. *Implicit Weights Attached by the OLS TWFE Estimator*

In this section, we follow de Chaisemartin and D'Haultfœuille (2020) to compute the implicit weights attached by the OLS TWFE estimator to individual cohort-year cells. We proceed in two steps. First, we reproduce our main results from columns 1 and 2 of Table 1 with the OLS estimator to ensure that they are comparable to the PPML results. For the baseline sample, the OLS TWFE point estimate (0.172) is very close to the corresponding PPML result (0.166), while the OLS ETWFE point estimate (0.347) is only slightly smaller than the corresponding PPML result (0.381). See also Section A.3.4 in the supplementary online Appendix for details.

Second, we follow de Chaisemartin and D'Haultfœuille (2020) to decompose the OLS TWFE estimate into cohort-year-specific treatment effects (from the OLS ETWFE estimation) and implicit weights attached to these cohort-year cells. By construction, the corresponding weighted sum of the cohort-year effects equals the OLS TWFE estimate. The results are presented in Figure 4, where we report the weights used in the computation of the aggregate treatment effects of the ETWFE from Section IB in dark color ("ETWFE") along with implicit weights attached by the OLS TWFE estimator to cohort-year cells computed following de Chaisemartin and D'Haultfœuille (2020) in light color ("TWFE").

Figure 4, panel A shows the results aggregated by event time. The aggregation scheme for the target parameter in Section IB gives equal weight to every posttreatment observation. As a result, the weights of the ETWFE estimator decrease by event time as the number of cohorts with data for the corresponding number of posttreatment years declines toward the end of the sample. In comparison, the

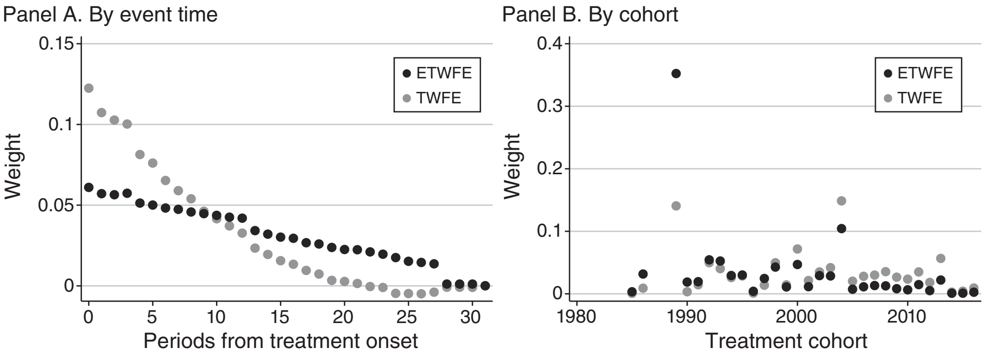

Figure 4. Weights of OLS ETWFE and TWFE Estimator

Notes: The figure reports the weights used in the computation of the aggregate treatment effects of the ETWFE estimator from Section IB in dark color ("ETWFE") along with the implicit weights attached by the OLS TWFE estimator to cohort-year cells computed following de Chaisemartin and D'Haultfœuille (2020) in light color ("TWFE"). Panel A reports weights aggregated by event time. Panel B reports weights aggregated by cohort.

implicit weights of the TWFE estimator are more than two times larger in the first treatment year and then decline much more rapidly. After ten years, the weights of the TWFE estimator are lower than those of the ETWFE estimator. They drop to zero at around 20 years after treatment onset and become slightly negative thereafter. In sum, the TWFE estimator seems to substantially overweight treatment effects shortly after treatment onset, which tend to be small, and to substantially underweight treatment effects further from treatment onset, which tend to be large due to maturation effects (Figure 3, panel A).

Figure 4, panel B shows the results aggregated by cohort. From this perspective, the TWFE estimator overweights late-treated cohorts (with small effects) and underweights early-treated cohorts (with large effects). Looking at weights by event time for different cohorts confirms that early (late) cohorts are more strongly underweighted (overweighted) in all years, which suggests that the cohort effect is not only mechanically driven by compositional differences in terms of event years (see Figure A1 in the supplementary online Appendix).

Overall, around 13 percent (60 out of 469) of all cohort-year cells have negative weights summing to a total of -0.0284. Therefore, while negative weights are present in this setting, they do not seem to play an important role for driving the differences between the TWFE and ETWFE estimates. Rather, in line with the example in Section IB, the TWFE is biased downward relative to the target parameter of the ETWFE estimate by putting larger weights on treatment effects shortly after treatment onset and on late cohorts, which are associated with small treatment effects.

IV. Robustness Analysis

In order to highlight the practical importance of the proposed methods and to demonstrate the robustness of our main results, we perform a battery of sensitivity experiments. To ease exposition and add some structure to the analysis, we split the robustness checks into four groups covering (i) the degree of heterogeneity of the estimator, (ii) potential IPPs, (iii) DiD methods, and (iv) gravity-related experiments. For brevity, we only provide a brief list of our experiments and a short summary and conclusion; a detailed discussion of the design and results from these additional analyses may be found in the supplementary online Appendix.

First, with regard to the degree of heterogeneity of the ETWFE estimator, as a robustness check, we impose restrictions on the treatment effect heterogeneity in the estimation, that is, the coefficient δ_{gs} in equation (3). In addition, we consider a more flexible specification, which includes an interaction of the cohort dummy with an indicator for individual agreements.[31] Relatedly, an imputation estimator obtains noisy (yet not consistent) estimates of individual treatment effects, which can then be used to compute more aggregated average treatment effects (Borusyak, Jaravel, and Spiess forthcoming). While the imputation estimator and the extended TWFE approach are generally not the same for nonlinear DiD, as discussed in Wooldridge (2023), they yield numerically equivalent treatment effects when the canonical link function is in the linear exponential family as in the case under consideration. However, additional estimations show that the equivalence seems to break down under the more complex fixed effect structure commonly used in the gravity setting. Most importantly for current purposes, the imputation estimate is still around twice as large as the associated TWFE estimate. See Section A.3.1 in the supplementary online Appendix for details.

Second, we consider potential IPPs, which may arise in nonlinear models with fixed effects.[32] A Monte Carlo simulation following Weidner and Zylkin (2021) suggests that, given the relatively large time dimension considered in our setting, IPP might be less of a problem for the coefficient estimate of the ETWFE, while the associated standard error is potentially downward biased. As a potential remedy, we consider a version of the (split-panel) jackknife studied in Dhaene and Jochmans (2015), Pfaffermayr (2021), and Weidner and Zylkin (2021) for bias correction. Simulation results suggest that the jackknife might help with any remaining bias of the ETWFE estimator and also exhibit approximately correct coverage probabilities. See Section A.3.2 in the supplementary online Appendix for details.

Third, we perform five sets of experiments based on developments in the DiD literature. First, we provide some additional robustness checks of our results with regard to the degree of heterogeneity of the ETWFE estimator by using binned

[31] We note that—similar to other estimators proposed in the literature (e.g., de Chaisemartin and D'Haultfœuille 2020; Callaway and Sant'Anna 2021; Sun and Abraham 2021)—while the model in equation (3) only allows identification of (simple) average treatment effects at the cohort-period level, it does not require treatment effects to be homogeneous within cohort-period cells as long as the estimation target is a weighted sum of cohort-period effects as considered in our paper (and the cited papers). However, if interest lies in different estimation targets that require not just simple average treatment effects of cohort-year cells but more granular treatment effects, then one needs to allow for "more heterogeneity" ex ante in the model by adding additional interactions such as an agreement indicator.

[32] This point has attracted significant and ongoing attention in the related trade literature, where the current consensus seems to be that the PPML estimator with TWFE is asymptotically unbiased even when the time dimension of the panel is fixed (Fernández-Val and Weidner 2016), while the three-way PPML estimator may be asymptotically biased for small T due to the IPP (Weidner and Zylkin 2021). Furthermore, estimates of cluster-robust sandwich-type standard errors may be downward biased in both two-way and three-way gravity settings (Jochmans 2017; Pfaffermayr 2021; Weidner and Zylkin 2021).

intervals and binned endpoints. Second, we restrict the control group to include only the never-treated group or the not-yet-treated group. Third, we consider an extension of the ETWFE estimation approach where the cohort-time-specific treatment effects are allowed to vary by time-constant covariates (e.g., bilateral distance, contiguity, common language, and past colonial relations). Fourth, we test the robustness of our results to the choice of the treatment onset. Instead of assuming an "onset" of RTAs three years before their entry into force, we omit these time periods in treated country pairs following Wooldridge (2023). Lastly, we experiment with alternative weighting schemes, in which we give every cohort, every event year, or every cohort-year the same weight to reduce the impact of sample composition and limit the influence of individual agreements on the aggregate result. See Section A.3.3 in the supplementary online Appendix for details.

Fourth, we also perform nine additional robustness experiments that are motivated by (historical) developments in the trade gravity literature. First, instead of PPML, we use the OLS estimator (see also Section IIID). Second, instead of consecutive-year data, we use five-year-interval data. Third, we estimate the model without data on domestic trade flows. Fourth, we introduce GATT/WTO membership as an additional covariate to our main specification. Fifth, we replace the country-pair fixed effects in our model with a set of "standard" gravity variables, including the log of bilateral distance, and dummy variables for sharing common borders, common language, and colonial ties. Sixth, we estimate a specification that does not include exporter-time and importer-time fixed effects and, instead, uses some observable country-specific covariates. Seventh, we explore the importance of the zero trade flows in our sample. Eight, we investigate the robustness of our results to alternative clusterings of the standard errors. Lastly, we allow for heterogeneous effects of "deep" versus "shallow" RTAs. See Section A.3.4 in the supplementary online Appendix for details.

In conclusion, our main result that the ETWFE RTA effects are significantly larger than previously thought based on TWFE estimates is confirmed and reinforced based on all of the robustness experiments that we have performed. For further details on these experiments, we refer the reader to the supplementary online Appendix and to Nagengast and Yotov (2023).

V. Conclusion

Motivated by the widely heterogeneous (across group and across time periods) policy estimates in the trade gravity literature and the concerns from recent econometric papers that TWFE estimates may be biased in staggered DiD designs—where the intervention occurs in multiple units in different time periods—we nest an extended TWFE estimator à la Wooldridge (2023) within the empirical structural gravity model. To test the implications of the resulting methods, we estimate the impact of one of the most widely studied trade policies—RTAs.

The main results from our analysis are that the ETWFE estimator delivers RTA estimates that are significantly larger and more long-lasting than corresponding estimates that are based on the current TWFE methods from the gravity literature. On a related note, the larger ETWFE estimates of the effects of RTAs are also more plausible from a policy perspective. Computing the implicit weights attached by the (OLS) TWFE to individual cohort-year effects following de Chaisemartin and D'Haultfœuille (2020) suggests that the TWFE estimate is biased downward relative to the target parameter of the ETWFE estimate by putting larger weights on short-run effects and on those of late cohorts, which are both associated with smaller treatment effects. A series of sensitivity experiments confirms the robustness of our main findings, in particular, to the heterogeneity of the ETWFE estimator and potential IPPs, as well as to various specifications from the DiD and gravity literatures.

Based on our analysis of the impact of RTAs, we expect that the ETWFE estimator (and, in the future, related heterogeneity-robust staggered DiD methods generalized to the nonlinear setting) may have significant implications for the estimates of other policies that have been studied with the gravity model of trade, for example, economic sanctions, membership in GATT and WTO, currency unions, etc., which all share the common feature that the treatment or intervention occurs in multiple units and in different time periods and is likely characterized by a large degree of treatment effect heterogeneity. Moreover, we expect that the new heterogeneity-robust staggered DiD methods may have significant implications for many of the policy estimates in gravity models of migration, FDI, and technology and innovation flows. Beyond the binary treatment considered in this paper, studying treatment effect heterogeneity for continuous treatments such as tariffs in the nonlinear gravity settings is a promising area of future research.[33]

REFERENCES

Agnosteva, Delina E., James E. Anderson, and Yoto V. Yotov. 2019. "Intra-national Trade Costs: Assaying Regional Frictions." *European Economic Review* 112: 32–50.

Anderson, James E. 2011. "The Gravity Model." *Annual Review of Economics* 3: 133–60.

Anderson, James E., and Eric van Wincoop. 2003. "Gravity with Gravitas: A Solution to the Border Puzzle." *American Economic Review* 93 (1): 170–92.

Anderson, James E., and Yoto V. Yotov. 2016. "Terms of Trade and Global Efficiency Effects of Free Trade Agreements, 1990–2002." *Journal of International Economics* 99: 279–98.

Anderson, James E., Mario Larch, and Yoto V. Yotov. 2018. "GEPPML: General Equilibrium Analysis with PPML." *World Economy* 41 (10): 2750–82.

Anderson, James E., Mario Larch, and Yoto V. Yotov. 2019. "Trade and Investment in the Global Economy: A Multi-country Dynamic Analysis." *European Economic Review* 120: 103311.

Athey, Susan, and Guido W. Imbens. 2006. "Identification and Inference in Nonlinear Difference-in-Differences Models." *Econometrica* 74 (2): 431–97.

Athey, Susan, and Guido W. Imbens. 2022. "Design-Based Analysis in Difference-In-Differences Settings with Staggered Adoption." *Journal of Econometrics* 226 (1): 62–79.

Baier, Scott L., and Jeffrey H. Bergstrand. 2007. "Do Free Trade Agreements Actually Increase Members' International Trade?" *Journal of International Economics* 71 (1): 72–95.

Baier, Scott L., and Jeffrey H. Bergstrand. 2021. "NSF-Kellogg Institute Data Base on Economic Integration Agreements." Kellogg Institute. https://kellogg.nd.edu/nsf-kellogg-institute-data-base-economic-integration-agreements (accessed October 15, 2024).

Baier, Scott L., Yoto V. Yotov, and Thomas Zylkin. 2019. "On the Widely Differing Effects of Free Trade Agreements: Lessons from Twenty Years of Trade Integration." *Journal of International Economics* 116: 206–26.

Baker, Andrew C., David F. Larcker, and Charles C. Y. Wang. 2022. "How Much Should We Trust Staggered Difference-in-Differences Estimates?" *Journal of Financial Economics* 144 (2): 370–95.

[33] See, for example, de Chaisemartin et al. (2022) for the linear case.

Baldwin, Richard E., and Daria Taglioni. 2006. "Gravity for Dummies and Dummies for Gravity Equations." NBER Working Paper 12516.

Bergstrand, Jeffrey H., Mario Larch, and Yoto V. Yotov. 2015. "Economic Integration Agreements, Border Effects, and Distance Elasticities in the Gravity Equation." *European Economic Review* 78: 307–27.

Borusyak, Kirill, and Xavier Jaravel. 2017. "Revisiting Event Study Design." Harvard University Working Paper.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. Forthcoming. "Revisiting Event Study Designs: Robust and Efficient Estimation." *Review of Economic Studies*.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–30.

Carrère, Céline, Monika Mrázová, and J. Peter Neary. 2020. "Gravity without Apology: The Science of Elasticities, Distance, and Trade." *Economic Journal* 130 (628): 880–910.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134 (3): 1405–54.

Ciani, Emanuele, and Paul Fisher. 2019. "Dif-in-Dif Estimators of Multiplicative Treatment Effects." *Journal of Econometric Methods* 8 (1): 20160011.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–96.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2024. "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *The Review of Economics and Statistics*. https://doi.org/10.1162/rest_a_01414.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2023. "Two-Way Fixed Effects and Differences-in-Differences with Heterogeneous Treatment Effects: A Survey." *Econometrics Journal* 26 (3): C1–C30.

de Chaisemartin, Clément, Xavier D'Haultfœuille, Félix Pasquier, Doulo Sow, and Gonzalo Vazquez-Bare. 2022. "Difference-in-Differences Estimators for Treatments Continuously Distributed at Every Period." arXiv: 2201.06898.

Deshpande, Manasi, and Yue Li. 2019. "Who Is Screened Out? Application Costs and the Targeting of Disability Programs." *American Economic Journal: Economic Policy* 11 (4): 213–48.

Dhaene, Geert, and Koen Jochmans. 2015. "Split-Panel Jackknife Estimation of Fixed-Effect Models." *Review of Economic Studies* 82 (3): 991–1030.

Eaton, Jonathan, and Samuel Kortum. 2002. "Technology, Geography, and Trade." *Econometrica* 70 (5): 1741–79.

Egger, Peter, and Mario Larch. 2008. "Interdependent Preferential Trade Agreement Memberships: An Empirical Analysis." *Journal of International Economics* 76 (2): 384–99.

Egger, Peter H., and Sergey Nigai. 2016. "World-Trade Growth Accounting." CESifo Working Paper 5831.

Egger, Peter H., and Filip Tarlea. 2015. "Multi-way Clustering Estimation of Standard Errors in Gravity Models." *Economics Letters* 134 (C): 144–47.

Egger, Peter H., Marko Koethenbuerger, and Gabriel Loumeau. 2022. "Local Border Reforms and Economic Activity." *Journal of Economic Geography* 22 (1): 81–102.

Egger, Peter H., Mario Larch, and Yoto V. Yotov. 2022. "Gravity Estimations with Interval Data: Revisiting the Impact of Free Trade Agreements." *Economica* 89 (353): 44–61.

Felbermayr, Gabriel, Aleksandra Kirilakha, Constantinos Syropoulos, Erdal Yalcin, and Yoto V. Yotov. 2020. "The Global Sanctions Data Base." *European Economic Review* 129: 103561.

Fernández-Val, Iván, and Martin Weidner. 2016. "Individual and Time Effects in Nonlinear Panel Models with Large N, T." *Journal of Econometrics* 192 (1): 291–312.

Gardner, John. 2022. "Two-Stage Differences in Differences." arXiv: 2207.05943.

Gurevich, Tamara, and Peter Herman. 2018. "The Dynamic Gravity Dataset: 1948–2016." USITC Working Paper 2018-02-A.

Heckman, James J., Hidehiko Ichimura, and Petra E. Todd. 1997. "Matching as an Econometric Evaluation Estimator: Evidence from Evaluating a Job Training Programme." *Review of Economic Studies* 64 (4): 605–54.

Hufbauer, Gary Clyde, and Barbara Oegg. 2003. *The Impact of Economic Sanctions on US Trade: Andrew Rose's Gravity Model*. Policy Briefs PB03-04. Washington, DC: Peterson Institute for International Economics.

Hummels, David. 2001. "Toward a Geography of Trade Costs." Unpublished. https://docs.lib.purdue.edu/gtapwp/17/.

Jochmans, Koen. 2017. "Two-Way Models for Gravity." *Review of Economics and Statistics* 99 (3): 478–85.

Larch, Mario, and Yoto V. Yotov. 2023. "Estimating the Effects of Trade Agreements: Lessons From 60 Years of Methods and Data." Drexel University School of Economics Working Paper 2023-4.

Larch, Mario, José-Antonio Monteiro, Roberta Piermartini, and Yoto V. Yotov. 2019b. "On the Effects of GATT/WTO Membership on Trade: They are Positive and Large after All." Drexel University School of Economics Working Paper 2019-4.

Larch, Mario, Joschka Wanner, Yoto V. Yotov, and Thomas Zylkin. 2019a. "Currency Unions and Trade: A PPML Re-assessment with High-Dimensional Fixed Effects." *Oxford Bulletin of Economics and Statistics* 81 (3): 487–510.

Moser, Christoph, and Andrew K. Rose. 2012a. "Why Do Trade Negotiations Take So Long?" *Journal of Economic Integration* 27 (2): 280–90.

Moser, Christoph, and Andrew K. Rose. 2012b. "Why Do Trade Negotiations Take So Long?" *VoxEU*, June 8. https://cepr.org/voxeu/columns/why-do-trade-negotiations-take-so-long.

Moser, Christoph, and Andrew K. Rose. 2014. "Who Benefits from Regional Trade Agreements? The View from the Stock Market." *European Economic Review* 68: 31–47.

Nagengast, Arne, and Yoto Yotov. 2023. "Staggered Difference-in-Differences in Gravity Settings: Revisiting the Effects of Trade Agreements." Drexel University School of Economics Working Paper 2023-6.

Nagengast, Arne J., and Yoto V. Yotov. 2025. *Data and Code for: "Staggered Difference-in-Differences in Gravity Settings: Revisiting the Effects of Trade Agreements."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. https://doi.org/10.3886/E194561V1.

Olivero, María Pía, and Yoto V. Yotov. 2012. "Dynamic Gravity: Endogenous Country Size and Asset Accumulation." *Canadian Journal of Economics* 45 (1): 64–92.

Persyn, Damiaan, and Wouter Torfs. 2016. "A Gravity Equation for Commuting with an Application to Estimating Regional Border Effects in Belgium." *Journal of Economic Geography* 16 (1): 155–75.

Pfaffermayr, Michael. 2021. "Confidence Intervals for the Trade Cost Parameters of Cross-Section Gravity Models." *Economics Letters* 201: 109787.

Rios-Avila, Fernando. 2022. "JWDID: Stata Module to Estimate Difference-in-Difference Models Using Mundlak Approach." Statistical Software Components, Boston College Department of Economics. https://ideas.repec.org/c/boc/bocode/s459114.html (accessed October 17, 2022).

Rose, Andrew K. 2000. "One Money, One Market: The Effect of Common Currencies on Trade." *Economic Policy* 15 (30): 8–45.

Rose, Andrew K. 2004. "Do We Really Know That the WTO Increases Trade?" *American Economic Review* 94 (1): 98–114.

Roth, Jonathan. 2024. "Interpreting Event-Studies from Recent Difference-in-Differences Methods." arXiv:2401.12309.

Roth, Jonathan, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." *Journal of Econometrics* 235 (2): 2218–44.

Santos Silva, João M. C., and Silvana Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88 (4): 641–58.

Santos Silva, João M. C., and Silvana Tenreyro. 2021. "The Log of Gravity At 15." University of Surrey Economics Discussion Paper 0121.

Schmidheiny, Kurt, and Sebastian Siegloch. 2023. "On Event Studies and Distributed-Lags in Two-Way Fixed Effects Models: Identification, Equivalence, and Generalization." Journal of Applied Economics 38(5):695–713.

Shahn, Zach. 2023. "Subgroup Difference in Differences to Identify Effect Modification without a Control Group." arXiv: 2306.11030.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225 (2): 175–99.

Teti, Feodora. 2020. "30 Years of Trade Policy: Evidence from 5.7 Billion Tariffs." ifo Working Paper 334.

Tinbergen, Jan. 1962. *Shaping the World Economy: Suggestions for an International Economic Policy*. New York, NY: Twentieth Century Fund.

United States International Trade Commission. 2021a. "Dynamic Gravity Dataset 1948–1999." United States International Trade Commission. https://www.usitc.gov/data/gravity/dgd.htm (accessed October 22, 2022).

United States International Trade Commission. 2021b. "Dynamic Gravity Dataset 2000–2019." United States International Trade Commission. https://www.usitc.gov/data/gravity/dgd.htm (accessed October 22, 2022).

Weidner, Martin, and Thomas Zylkin. 2021. "Bias and Consistency in Three-Way Gravity Models." *Journal of International Economics* 132: 103513.

Wooldridge, Jeffrey M. 2021. "Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators." http://dx.doi.org/10.2139/ssrn.3906345.

Wooldridge, Jeffrey M. 2023. "Simple Approaches to Nonlinear Difference-in-Differences with Panel Data." *Econometrics Journal* 26 (3): C31–C66.

World Trade Organization. 2019. "Structural Gravity Database." World Trade Organization. https://www.wto.org/english/res_e/reser_e/structural_gravity_e.htm (accessed October 23, 2022).

Yotov, Yoto V. 2022. "On the Role of Domestic Trade Flows for Estimating the Gravity Model of Trade." *Contemporary Economic Policy* 40 (3): 526–40.

Yotov, Yoto V., Roberta Piermartini, José-Antonio Monteiro, and Mario Larch. 2016. *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model*. Geneva, Switzerland: United Nations and World Trade Organization. https://www.wto.org/english/res_e/booksp_e/advancedwtounctad2016_e.pdf.